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METASURFACE REFLECTOR, PROJECTION DEVICE, NEAR-EYE WEARABLE DEVICE, AND METHOD FOR MANUFACTURING METASURFACE REFLECTOR

Abstract

A metasurface reflector includes: a first metal layer and a second metal layer stacked in a first direction; a dielectric layer provided between the first metal layer and the second metal layer in the first direction; and a protective layer covering the second metal layer. The metasurface reflector is divided into unit regions arranged in a second direction along a main surface of the dielectric layer and in a third direction along the main surface and intersecting the second direction. The second metal layer includes metal units respectively provided in all or some of the unit regions. The protective layer is made of a metal having a standard electrode potential higher than that of a metal constituting the second metal layer. The protective layer includes a first portion covering a top surface of each metal unit and a second portion covering a side surface of each metal unit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the benefit of priority from Japanese Patent Application No. 2024-33223 filed with the Japan Patent Office on Mar. 5, 2024, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a metasurface reflector, a projection device, a near-eye wearable device, and a method for manufacturing a metasurface reflector.

BACKGROUND

[0003] Reflectors using metasurface technology are known. For example, US Patent Application Publication No. 2018/0113310 describes a near-eye display assembly including an image source and a combiner including a nanostructured surface optically coupled to the image source. The unit cell of the nanostructure surface is configured by stacking a base layer, a dielectric layer, and a meta atom layer in that order.

SUMMARY

[0004] In the near-eye display assembly described in US Patent Application Publication No. 2018/0113310, the meta atom layer is exposed. When a metal layer such as silver and aluminum is used as the meta atom layer, the metal layer may be oxidized or sulfurized when exposed to air. In this case, the optical characteristics may be changed and the reflection characteristics may be deteriorated.

[0005] The present disclosure describes a metasurface reflector, a projection device, a near-eye wearable device, and a method for manufacturing a metasurface reflector capable of suppressing deterioration of reflection characteristics.

[0006] A metasurface reflector according to one aspect of the present disclosure includes: a first metal layer and a second metal layer stacked in a first direction; a dielectric layer provided between the first metal layer and the second metal layer in the first direction; and a protective layer covering the second metal layer. The dielectric layer includes a main surface on which the second metal layer is provided. The metasurface reflector is divided into a plurality of unit regions arranged in a second direction along the main surface and in a third direction along the main surface and intersecting the second direction. The second metal layer includes metal units respectively provided in all or some of the plurality of unit regions. The protective layer is made of a metal having a standard electrode potential higher than that of a metal constituting the second metal layer. Each of the metal units includes a bottom surface facing the dielectric layer in the first direction, a top surface provided opposite to the bottom surface in the first direction, and a side surface connecting the top surface and the bottom surface. The protective layer includes a first portion covering the top surface and a second portion covering the side surface.

[0007] In the metasurface reflector, the protective layer covers the top and side surfaces of the metal unit. Therefore, since the top and side surfaces of the metal unit are not exposed to air when the metasurface reflector is used, the possibility that the metal unit is oxidized and sulfurized is

reduced. Furthermore, since the protective layer is made of a metal having a standard electrode potential higher than that of a metal constituting the second metal layer, the protective layer is less likely to be oxidized and sulfurized than the second metal layer. Accordingly, in the metasurface reflector, a change in the optical characteristics hardly occurs. As a result, deterioration of the reflection characteristics can be suppressed.

[0008] A thickness of the protective layer may be 20% or less of a sum of a length of the second metal layer in the first direction and the thickness of the protective layer. In this case, the protective layer can be provided while suppressing the influence on the reflection characteristics of the metasurface reflector. Accordingly, deterioration of the reflection characteristics can be further suppressed.

[0009] Each of the metal units may be a metal body having a trapezoidal shape when viewed from the first direction. In this case, the structure of the metal unit can be simplified as compared with the case where the metal unit is composed of a plurality of metal bodies. Accordingly, the manufacturing of the metasurface reflector can be facilitated.

[0010] A length of the metal body in the second direction may be 500 nm or more and 2500 nm or less. A length of the metal body in the first direction may be 10 nm or more and 100 nm or less. A length of a short side of the metal body may be 10 nm or more and 200 nm or less. A length of a long side of the metal body may be larger than the length of the short side and may be 100 nm or more and 500 nm or less. In this case, the reflection efficiency for visible light can be enhanced.

[0011] The protective layer may be made of a metal containing at least one element selected from a group consisting of gold, ruthenium, and iridium. In this case, the deterioration of the reflection characteristics can be suppressed as compared with molybdenum, titanium, tungsten and the like.

[0012] The second metal layer may be made of a metal containing at least one element selected from a group consisting of silver, aluminum, and copper. In this case, the second metal layer having a relatively high reflectance and a relatively high electric conductivity can be obtained.

Accordingly, the electromagnetic resonance between the first metal layer and the second metal layer can be strengthened, and the reflection efficiency can be enhanced.

[0013] The dielectric layer may be made of a material transparent in a visible light region. In this case, since the absorption rate of visible light in the dielectric layer is suppressed, the reflection efficiency for visible light can be enhanced.

[0014] The dielectric layer may be made of a compound selected from a group consisting of silicon oxide, titanium oxide, magnesium oxide, and aluminum oxide. In this case, the dielectric layer having a dielectric constant that does not interfere with the electromagnetic action can be obtained. Accordingly, the electromagnetic resonance between the first metal layer and the second metal layer can be strengthened, and the reflection efficiency can be enhanced.

[0015] A length of the dielectric layer in the first direction may be 10 nm or more and 100 nm or less. A length of the first metal layer in the first direction may be 50 nm or more and 1000 nm or less. In this case, the possibility that the dielectric layer interferes with the electromagnetic action can be reduced, and the possibility that the laser light passes through the first metal layer can be reduced. Accordingly, the reflection efficiency can be enhanced.

[0016] A thickness of the second portion may decrease as a distance from the dielectric layer increases in the first direction. The thinner the second portion is, the higher the reflected electric field strength becomes. Therefore, the protective layer can be provided while suppressing the influence on the reflection characteristics of the metasurface reflector. Therefore, the deterioration of the reflection characteristics can be further suppressed.

[0017] A projection device according to another aspect of the present disclosure is a projection device mounted on a near-eye wearable device. The projection device includes: a light source that emits laser light; a movable mirror that performs scanning with the laser light; and the above-described metasurface reflector that reflects the laser light having passed through the movable mirror to cause a user wearing the near-eye wearable device to visually recognize an image. In the

projection device as well, deterioration of the reflection characteristics can be suppressed.

[0018] A near-eye wearable device according to still another aspect of the present disclosure includes: the above-described projection device; and a lens provided with the metasurface reflector. In the near-eye wearable device as well, deterioration of the reflection characteristics can be suppressed.

[0019] A method for manufacturing a metasurface reflector according to still another aspect of the present disclosure includes: a step of preparing a laminate in which a first metal layer, a dielectric layer, a second metal layer, and a third metal layer are sequentially laminated in a first direction; a step of forming a resist film on the third metal layer; a step of forming, on the resist film, a pattern for forming a metal unit constituting a metasurface reflector, the metal unit including a bottom surface facing the dielectric layer in the first direction, a top surface provided opposite to the bottom surface in the first direction, and a side surface connecting the top surface and the bottom surface; a step of forming the metal unit and a first protective film covering the top surface by milling the second metal layer and the third metal layer using the pattern; and a step of forming a second protective film covering the side surface of the metal unit. The first protective film and the second protective film are made of a metal having a standard electrode potential higher than that of a metal constituting the second metal layer.

[0020] In the method for manufacturing a metasurface reflector, the pattern is formed in the resist film formed on the third metal layer of the laminate, and the metal unit and the first protective film covering the top surface of the metal unit are formed by milling the second metal layer and the third metal layer using the pattern. Further, the second protective film is formed on the side surface of the metal unit. In the metasurface reflector manufactured in this manner, since the top surface and the side surface of the metal unit are not exposed to the air when the metasurface reflector is used, the possibility that the metal unit is oxidized and sulfurized is reduced. Furthermore, since the first protective film and the second protective film are made of a metal having a standard electrode potential higher than that of the metal constituting the second metal layer, the first protective film and the second protective film are less likely to be oxidized and sulfurized than the second metal layer. Accordingly, in the metasurface reflector, the optical characteristics hardly change. As a result, deterioration of the reflection characteristics can be suppressed.

[0021] The step of forming the second protective film may include: a step of removing the resist film; a step of forming a metal film on a surface of a structure obtained by removing the resist film; and a step of forming the second protective film by removing a portion of the metal film formed on a main surface of the dielectric layer. In this case, the second protective film can be formed on the side surface of the metal unit by a relatively simple process.

[0022] In the step of forming the resist film, a multilayer resist film including a lower resist film and an upper resist film may be formed on the third metal layer. The step of forming the second protective film may include: a step of forming a metal film on a surface of a structure obtained by the step of forming the first protective film; a step of forming the second protective film by removing a portion of the metal film formed on a main surface of the dielectric layer; and a step of removing the multilayer resist film after the second protective film is formed. In this case, the multilayer resist film is removed in a state where the top surface of the metal unit is covered with the first protective film and the side surface of the metal unit is covered with the second protective film. Therefore, since the top surface and the side surface of the metal unit are not exposed to the solvent used for removing the multilayer resist film, the possibility that the metal unit is corroded is reduced. As a result, deterioration of the reflection characteristics of the metasurface reflector can be further suppressed.

[0023] In the step of forming the metal film, the metal film may be formed using glancing angle sputtering. In this case, the metal film is easily formed on the side surface of the metal unit. Therefore, the time required for manufacturing the metasurface reflector can be reduced.

[0024] According to each aspect and each embodiment of the present disclosure, deterioration of the reflection characteristics can be suppressed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] FIG. 1 is a perspective view showing an appearance of a near-eye wearable device to which a metasurface reflector according to an embodiment is applied.

[0026] FIG. 2 is a configuration diagram schematically showing the projection device shown in FIG. 1.

[0027] FIG. 3 is an enlarged view of the metasurface reflector shown in FIG. 2.

[0028] FIG. 4 is a perspective view schematically showing the unit region shown in FIG. 3.

[0029] FIG. 5 is a cross-sectional view taken along the line V-V of FIG. 4.

[0030] FIG. 6 is a diagram for explaining the principle of reflection by the metasurface reflector shown in FIG. 2.

[0031] FIG. 7 is a diagram showing the phase change amount of reflected light at the position of the metasurface reflector in the X-axis direction.

[0032] FIG. 8 is a diagram for explaining the relationship between the position in the X-axis direction and the length of the metal unit in the X-axis direction.

[0033] FIG. 9 is a diagram showing the relationship between the proportion of the protective layer and the reflected electric field strength.

[0034] FIG. 10 is a flowchart showing an example of a method for manufacturing a metasurface reflector.

[0035] FIG. 11 is a view for explaining the resist film forming step shown in FIG. 10.

[0036] FIG. 12 is a view for explaining the pattern forming step shown in FIG. 10.

[0037] FIG. 13 is a view for explaining the first protective film forming step shown in FIG. 10.

[0038] FIG. 14 is a view for explaining the resist film removal step shown in FIG. 10.

[0039] FIG. 15 is a view for explaining the metal film forming step shown in FIG. 10.

[0040] FIG. 16 is a view for explaining the metal film removal step shown in FIG. 10.

[0041] FIG. 17 is a flowchart showing another example of a method for manufacturing a metasurface reflector.

[0042] FIG. 18 is a view for explaining the resist film forming step shown in FIG. 17.

[0043] FIG. 19 is a view for explaining the pattern forming step shown in FIG. 17.

[0044] FIG. 20 is a view for explaining the first protective film forming step shown in FIG. 17.

[0045] FIG. 21 is a view for explaining the metal film forming step shown in FIG. 17.

[0046] FIG. 22 is a view for explaining the metal film removal step shown in FIG. 17.

[0047] FIG. 23 is a view for explaining the resist film removal step shown in FIG. 17.

DETAILED DESCRIPTION

[0048] Hereinafter, embodiments of the present disclosure will be described in detail with reference to the drawings. In the description of the drawings, the same elements are denoted by the same reference numerals, and redundant description will be omitted. In each figure, an XYZ coordinate system may be shown. The Y-axis direction (third direction) is a direction intersecting (for example, orthogonal to) the X-axis direction (second direction) and the Z-axis direction (first direction). The Z-axis direction is a direction intersecting (for example, orthogonal to) the X-axis direction and the Y-axis direction. In the present specification, the numerical ranges indicated by “to” represent ranges that include the values described before and after “to” as the minimum and maximum values, respectively. The individually described upper and lower limit values can be combined arbitrarily.

[0049] A near-eye wearable device to which a metasurface reflector according to an embodiment is

applied will be described with reference to FIG. 1. FIG. 1 is a perspective view showing an appearance of a near-eye wearable device to which a metasurface reflector according to an embodiment is applied. The near-eye wearable device **1** shown in FIG. 1 is a device for superimposing an image on the field of view of the real world. The near-eye wearable device **1** is, for example, a head-mounted device, and may take the form of an eyeglass type, a goggle type, a hat type, a helmet type, or the like. Examples of the near-eye wearable device **1** include smart glasses such as augmented reality (AR) glasses, and mixed reality (MR) glasses. The near-eye wearable device **1** includes a frame **2**, a lens **3**, and a projection device **10**.

[0050] The frame **2** includes a pair of rims **2a**, a bridge **2b**, and a pair of temples **2c**. The rim **2a** is a portion for holding the lens **3**. The bridge **2b** is a portion connecting the pair of rims **2a**. The temple **2c** extends from the rim **2a** and is a portion to be put on an ear of a user. The frame **2** may be a rimless frame. The lens **3** has an inner surface **3a** (refer to FIG. 2) facing an eyeball E (refer to FIG. 6) of a user wearing the near-eye wearable device **1**.

[0051] In the present embodiment, the projection device **10** is a device for directly projecting (drawing) an image onto a retina RE (refer to FIG. 6) of a user wearing the near-eye wearable device **1**. The projection device **10** is mounted on the near-eye wearable device **1**. In the present embodiment, the near-eye wearable device **1** includes two projection devices **10** in order to project an image onto both the right and left retinas RE, but may include only one of the projection devices **10**.

[0052] Next, the projection device **10** will be described in detail with reference to FIG. 2. FIG. 2 is a configuration diagram schematically showing the projection device shown in FIG. 1. As shown in FIG. 2, the projection device **10** includes an optical engine **20** and a metasurface reflector **30**.

[0053] The optical engine **20** is a device that generates laser light Ls having a color and intensity corresponding to a pixel of an image to be projected onto the retina RE and emits the laser light Ls to the metasurface reflector **30**. The optical engine **20** is mounted on each temple **2c**. The optical engine **20** includes a light source unit **21** (light source), optical components **22**, a movable mirror **23**, a laser driver **24**, a mirror driver **25**, and a controller **26**.

[0054] The light source unit **21** emits laser light. As the light source unit **21**, for example, a full-color laser module is used. The light source unit **21** includes a red laser diode, a green laser diode, a blue laser diode, and a multiplexer that multiplexes laser lights emitted from laser diodes into one laser light. The light source unit **21** emits the multiplexed laser light. The multiplexed laser light contains a component having a wavelength of red (red component), a component having a wavelength of green (green component), and a component having a wavelength of blue (blue component). The light source unit **21** emits laser light having a color and intensity corresponding to a pixel of an image to be projected onto the retina RE.

[0055] The optical components **22** are components that optically process the laser light emitted from the light source unit **21**. In the present embodiment, the optical components **22** include a collimator lens **22a**, a slit **22b**, and a neutral density filter **22c**. The collimator lens **22a**, the slit **22b**, and the neutral density filter **22c** are arranged in this order along the optical path of the laser light. The optical components **22** may have other configurations.

[0056] The movable mirror **23** is an optical component for performing scanning with the laser light Ls. The movable mirror **23** is provided in a direction in which the laser light processed by the optical components **22** is emitted. The movable mirror **23** is configured to be swingable about an axis extending in the horizontal direction (X-axis direction) of the lens **3** and about an axis extending in the vertical direction (Y-axis direction) of the lens **3**, for example, and reflects the laser light while changing the angle in the X-axis direction and the Y-axis direction. As the movable mirror **23**, for example, a micro electro mechanical systems (MEMS) mirror is used.

[0057] The laser driver **24** is a driving circuit for driving the light source unit **21**. The laser driver **24** drives the light source unit **21** based on, for example, the intensity of the laser light and the temperature of the light source unit **21**. The mirror driver **25** is a driving circuit for driving the

movable mirror **23**. The mirror driver **25** swings the movable mirror **23** within a predetermined angle range and at a predetermined timing. The controller **26** is a device for controlling the laser driver **24** and the mirror driver **25**.

[0058] In the optical engine **20**, laser light having a color and intensity corresponding to a pixel of an image to be projected onto the retina RE is emitted from the light source unit **21**, passes through the optical components **22**, and is reflected by the movable mirror **23**. The laser light reflected by the movable mirror **23** is emitted to the metasurface reflector **30** as the laser light Ls.

[0059] The metasurface reflector **30** is an optical component that reflects the laser light Ls that has passed through the movable mirror **23** to cause a user wearing the near-eye wearable device **1** to visually recognize an image. No image is displayed on the metasurface reflector **30**. The metasurface reflector **30** is provided on the inner surface **3a** of the lens **3**.

[0060] Next, the configuration of the metasurface reflector **30** will be described with reference to FIGS. **3** to **5**. FIG. **3** is an enlarged view of the metasurface reflector shown in FIG. **2**. FIG. **4** is a perspective view schematically showing the unit region shown in FIG. **3**. FIG. **5** is a cross-sectional view taken along the line V-V of FIG. **4**.

[0061] As shown in FIG. **3**, the metasurface reflector **30** is divided into a plurality of unit regions **31**. The plurality of unit regions **31** is provided along the inner surface **3a** of the lens **3**. The plurality of unit regions **31** is arranged in a two-dimensional array in the horizontal direction (X-axis direction) and the vertical direction (Y-axis direction) of the lens **3**.

[0062] As shown in FIGS. **4** and **5**, the metasurface reflector **30** includes a metal layer **41** (first metal layer), a dielectric layer **42**, a metal layer **43** (second metal layer), and a protective layer **44** in sequence in the Z-axis direction.

[0063] The metal layer **41** is a base layer. The metal layer **41** is provided on the inner surface **3a** of the lens **3**. The metal layer **41** is made of a metal having high reflection characteristics in the visible light region. The metal layer **41** is made of, for example, a metal containing at least one element selected from the group consisting of gold (Au), copper (Cu), silver (Ag), and aluminum (Al). The length (thickness d1) of the metal layer **41** in the Z-axis direction may be any length as long as the metal layer **41** is capable of passing a resonance current and reflecting light. The thickness d1 is, for example, 50 nm to 1000 nm.

[0064] The dielectric layer **42** is a layer functioning as a spacer. The dielectric layer **42** is provided between the metal layer **41** and the metal layer **43** in the Z-axis direction. In the present embodiment, the dielectric layer **42** is provided on the metal layer **41**. The dielectric layer **42** has a main surface **42a** on which the metal layer **43** is provided. The dielectric layer **42** has a dielectric constant that does not interfere with the electromagnetic action of the metal layer **41** and the metal layer **43**. The dielectric layer **42** is made of a material that is transparent in the visible light region. The dielectric layer **42** may be made of a material having a high dielectric constant in order to achieve high reflection characteristics. The dielectric layer **42** is made of, for example, one compound selected from the group consisting of silicon oxides (e.g., SiO₂), titanium oxides (e.g., TiO₂), magnesium oxides (e.g., MgO), and aluminum oxides (e.g., Al₂O₃). The length (thickness d2) of the dielectric layer **42** in the Z-axis direction is, for example, 10 nm to 100 nm.

[0065] The metal layer **43** is a layer for exciting electromagnetic resonance together with the metal layer **41**. The metal layer **41** and the metal layer **43** are stacked in the Z-axis direction with the dielectric layer **42** interposed therebetween. In the present embodiment, the metal layer **43** is provided on the main surface **42a** of the dielectric layer **42**. The metal layer **43** is made of a metal having high reflection characteristics in the visible light region. The metal layer **43** is made of a metal containing at least one element selected from the group consisting of silver (Ag), aluminum (Al), and copper (Cu), for example.

[0066] The metal layer **43** includes a plurality of metal units **45**. The metal unit **45** is provided in each of the plurality of unit regions **31**. Each metal unit **45** is configured such that the phase change

amount ϕ of the reflected light L_r by the metal unit **45** changes linearly from one end **31a** (first end) to the other end **31b** (second end) in the X-axis direction of the unit region **31** in which the metal unit **45** is provided. Further, each metal unit **45** is configured such that the phase change amount ϕ of the reflected light L_r changes substantially by 360° (2π radians) from one end **31a** to the other end **31b**. The phase change amount ϕ of the reflected light L_r is an amount by which the phase of the reflected light L_r from the phase of the reflected light L_r at a certain length of the metal unit **45** in the Y-axis direction changes when the length of the metal unit **45** in the Y-axis direction is changed. Hereinafter, the length in the Y-axis direction may be referred to as “width”. [0067] In the present embodiment, each metal unit **45** is a single metal body having a trapezoidal shape when viewed from the Z-axis direction. The length (thickness d_3) of each metal unit **45** in the Z-axis direction is, for example, 10 nm to 100 nm. The length of each metal unit **45** in the X-axis direction is equal to or slightly shorter than the length L_x of the unit region **31** in the X-axis direction. The length of each metal unit **45** in the X-axis direction is, for example, 500 nm to 2500 nm.

[0068] The length (width W_1) of the short side of each metal unit **45** is set, for example, to a value close to the resolution of the exposure device used for forming the metal unit **45**. The width W_1 is, for example, 10 nm to 200 nm. The length (width W_2) of the long side of each metal unit **45** is larger than the width W_1 , and is set to a length that a phase difference of substantially 360° (2π radians) can be obtained from the phase of the reflected light L_r at the width W_1 . The width W_2 is, for example, 100 nm to 500 nm. Each metal unit **45** is formed by photolithography, for example. [0069] The metal unit **45** includes a top surface **45a**, a bottom surface **45b**, and a side surface **45c**. The bottom surface **45b** is a surface facing the dielectric layer **42** in the Z-axis direction. The top surface **45a** is provided opposite to the bottom surface **45b** in the Z-axis direction. The side surface **45c** is a surface connecting the top surface **45a** and the bottom surface **45b**. The side surface **45c** connects the entire periphery of the top surface **45a** and the entire periphery of the bottom surface **45b**.

[0070] The protective layer **44** is a layer for protecting the metal layer **43** (metal unit **45**). The protective layer **44** is provided so as to cover the entirety of each metal unit **45**. That is, the metal unit **45** is not exposed. The protective layer **44** is made of a metal which is less susceptible to oxidation and sulfurization and has higher corrosion resistance than the metal layer **43**. In other words, the protective layer **44** is made of a metal having a standard electrode potential higher than that of the metal constituting the metal layer **43**. The protective layer **44** is made of, for example, a metal containing at least one element selected from the group consisting of gold (Au), ruthenium (Ru), and iridium (Ir). In a combination of these metals and a metal (for example, silver) constituting the metal layer **43**, the attenuation of the near-field light is small.

[0071] In the present embodiment, the protective layer **44** includes a plurality of metal bodies that respectively cover the plurality of metal units **45**. The thickness d_4 of the protective layer **44** (metal body) is 20% or less of the thickness d_5 which is the sum of the thickness d_3 and the thickness d_4 . The thickness d_4 is, for example, 2.5 nm to 25 nm. The thickness d_4 of the protective layer **44** is not limited to the length in the Z-axis direction, but means the length (layer thickness) from the surface of the metal unit **45** to the surface of the protective layer **44**.

[0072] The protective layer **44** includes a first portion **44a** and a second portion **44b**. The first portion **44a** covers the top surface **45a** of the metal unit **45**. The second portion **44b** covers the side surface **45c** of the metal unit **45**. The thickness of the second portion **44b** may decrease as the distance from the dielectric layer **42** in the Z-axis direction increases.

[0073] An adhesion layer **46** (refer to FIG. 11) may be provided between the inner surface **3a** of the lens **3** and the metal layer **41**. An adhesion layer **47** (refer to FIG. 11) may be provided between the metal layer **41** and the dielectric layer **42**. An adhesion layer **48** (refer to FIG. 16) may be provided between the dielectric layer **42** and the metal layer **43**. Each of the adhesion layers **46** to **48** is a layer for enhancing the adhesion between the two layers. Each of the adhesion layers **46** to **48** is

made of, for example, chromium (Cr). The length (film thickness) of each of the adhesion layers **46** to **48** in the Z-axis direction is about 3 nm.

[0074] Next, a method for determining the length L_x as well as the principle of reflection by the metasurface reflector **30** will be described with reference to FIGS. 5 to 8. FIG. 6 is a diagram for explaining the principle of reflection by the metasurface reflector shown in FIG. 2. FIG. 7 is a diagram showing the phase change amount of reflected light at the position of the metasurface reflector in the X-axis direction. FIG. 8 is a diagram for explaining the relationship between the position in the X-axis direction and the length of the metal unit in the X-axis direction.

[0075] As shown in FIGS. 5 and 6, each unit region **31** is a nanostructure configured to reflect the laser light L_s at a reflection angle $\theta_{\text{sub.r}}$ corresponding to a position where the unit region **31** is provided when the laser light L_s is incident at an incident angle $\theta_{\text{sub.i}}$ corresponding to the position where the unit region **31** is provided. The reflection angle $\theta_{\text{sub.r}}$ of each unit region **31** is set so that the laser light L_s (reflected light L_r) reflected by each unit region **31** passes through the center of the pupil PP. Therefore, the incident angle $\theta_{\text{sub.i}}$ and the reflection angle $\theta_{\text{sub.r}}$ are determined by the position where the unit region **31** is provided. The unit region **31** is configured so that the incident angle $\theta_{\text{sub.i}}$ and the reflection angle $\theta_{\text{sub.r}}$ corresponding to the position where the unit region **31** is provided are obtained.

[0076] Here, the incident angle $\theta_{\text{sub.i}}$ is an angle formed by a normal line of a surface irradiated with the laser light L_s and an incident direction of the laser light L_s . The reflection angle $\theta_{\text{sub.r}}$ is an angle formed by a normal line of a surface irradiated with the laser light L_s and an emission direction of the reflected light L_r . In the plane including the laser light L_s and the reflected light L_r , when the reflected light L_r is emitted on the side opposite to the incident light (laser light L_s) with the normal line as a boundary, the reflection angle $\theta_{\text{sub.r}}$ is expressed by a positive value, and when the reflected light L_r is emitted on the same side as the incident light (laser light L_s) with the normal line as a boundary, the reflection angle $\theta_{\text{sub.r}}$ is expressed by a negative value.

[0077] For example, as shown in FIG. 6, when the pupil PP of the user faces the front, the unit regions **31** provided from the position Pa to the position Pc in the X-axis direction are used. The laser light L_s reflected by the unit region **31** provided at the position Pa corresponds to a pixel at the right end of the image. The position Pb is in the middle between the position Pa and the position Pc, and the laser light L_s reflected by the unit region **31** provided at the position Pb corresponds to a pixel in the center of the image. The laser light L_s reflected by the unit region **31** provided at the position Pc corresponds to a pixel at the left end of the image.

[0078] In the unit region **31** provided at the position Pa, the laser light L_s is incident at an incident angle $\theta_{\text{sub.i}}$ of 30° , and the laser light L_s is reflected at a reflection angle $\theta_{\text{sub.r}}$ of 5° to be emitted as the reflected light L_r . In the unit region **31** provided at the position Pb, the laser light L_s is incident at an incident angle $\theta_{\text{sub.i}}$ of 40° , and the laser light L_s is reflected at a reflection angle $\theta_{\text{sub.r}}$ of -5° to be emitted as the reflected light L_r . In the unit region **31** provided at the position Pc, the laser light L_s is incident at an incident angle $\theta_{\text{sub.i}}$ of 50° , and the laser light L_s is reflected at a reflection angle $\theta_{\text{sub.r}}$ of -10° to be emitted as the reflected light L_r .

[0079] As shown in FIG. 7, the width of the metal unit **45** increases from the width W_1 at the one end **31a** to the width W_2 at the other end **31b**. The phase change amount ϕ at each position of the metal unit **45** in the X-axis direction is substantially the same as the phase change amount ϕ caused by the square metal body having a side having the same length as the width at the position in the plan view. The larger the area of the square metal body in the plan view, the larger the phase change amount ϕ (phase delay amount) at that position. Accordingly, since the laser light L_s is reflected with different phase change amounts ϕ in accordance with the position in the X-axis direction, the wave front is formed by the interference between the reflected lights. That is, a plane wave having the gradient of the function $\phi(x)$ indicating the relationship between the position x in the X-axis direction and the phase change amount ϕ as the wave vector Φ is generated.

[0080] Here, as shown in FIG. 5, by generalizing the Snell's law, the Snell's law is expressed by

Equation (1) using the wave vector $k_{\text{sub}0}$ of the laser light L_s , the incident angle $\theta_{\text{sub}i}$, the reflection angle $\theta_{\text{sub}r}$, and the wave vector Φ .

[00001][Equation1] $k_0 \sin \theta_i + \Phi = k_0 \sin \theta_r$

[0081] The wave vector $k_{\text{sub}0}$ is expressed by $2\pi/\lambda$ using the wavelength λ of the laser light L_s . The wave vector Φ is expressed by $2\pi/L_x$ using the length L_x of the unit region **31** in the X-axis direction. By transforming Equation (1) using these relations, Equation (2) is obtained.

[00002][Equation2] $\sin \theta_r = \sin \theta_i + \frac{L_x}{\lambda} \quad (2)$

[0082] The length L_x of the unit region **31** is obtained by substituting the wavelength λ of the laser light L_s and the incident angle $\theta_{\text{sub}i}$ and the reflection angle $\theta_{\text{sub}r}$ of the laser light L_s corresponding to the position where the unit region **31** is provided into Equation (2). The laser light L_s contains a red component, a green component, and a blue component, but the length L_x is determined using, for example, the wavelength of the green component, to which the human eyes have the highest sensitivity, as the wavelength λ .

[0083] When the length L_x is a positive value, the shape of the metal unit **45** is set to a trapezoidal shape in which the width of the metal unit **45** increases from one end **31a** to the other end **31b**. When the length L_x is a negative value, the shape of the metal unit **45** is set to a trapezoidal shape in which the width of the metal unit **45** decreases from one end **31a** to the other end **31b**.

[0084] As described above, the length L_x of each unit region **31** is determined by the wavelength λ to be reflected, and the incident angle $\theta_{\text{sub}i}$ and the reflection angle $\theta_{\text{sub}r}$ corresponding to the position where the unit region **31** is provided. The length of the metal unit **45** in the X-axis direction is equal to or slightly shorter than the length L_x of the unit region **31** in the X-axis direction. Accordingly, the length of the metal unit **45** in the X-axis direction is determined by the wavelength λ to be reflected, and the incident angle $\theta_{\text{sub}i}$ and the reflection angle $\theta_{\text{sub}r}$ corresponding to the position of the unit region **31** where the metal unit **45** is provided.

[0085] When the reflection angle $\theta_{\text{sub}r}$ is smaller than the incident angle $\theta_{\text{sub}i}$, the length L_x becomes a negative value. As the difference between the incident angle $\theta_{\text{sub}i}$ and the reflection angle $\theta_{\text{sub}r}$ increases, the absolute value of the length L_x decreases. As shown in FIG. 8, in the near-eye wearable device **1**, as the distance from the movable mirror **23** increases, the reflection angle $\theta_{\text{sub}r}$ decreases, and the difference between the incident angle $\theta_{\text{sub}i}$ and the reflection angle $\theta_{\text{sub}r}$ increases, so that the length L_x also decreases. Accordingly, the lengths L_x of the unit regions **31** included in the same arrangement in the X-axis direction are different from each other, and the lengths in the X-axis direction of the metal units **45** included in the same arrangement in the X-axis direction are also different from each other.

[0086] Note that the length L_y of each unit region **31** is a predetermined fixed value. The length L_y is slightly larger than the width $W2$. The length L_y may be a length obtained by adding the resolution (for example, 100 nm) of the exposure device used for forming the metal unit **45** to the width $W2$, and is set to, for example, 600 nm. The width $W1$ and the width $W2$ of each metal unit **45** are predetermined fixed values. As described above, the width $W1$ is set to a value close to the resolution (for example, 100 nm) of the exposure device used for forming the metal unit **45**. The width $W2$ is set to a length (for example, 350 nm) at which a phase difference of substantially 360° (2π radians) can be obtained from the phase of the reflected light L_r at the width $W1$.

[0087] Next, the influence of the protective layer **44** on the intensity of the reflected light L_r will be described with reference to FIG. 9. FIG. 9 is a diagram showing the relationship between the proportion of the protective layer and the reflected electric field strength. The horizontal axis of FIG. 9 indicates the proportion (unit: %) of the protective layer **44**. The proportion of the protective layer **44** is represented by the proportion of the thickness $d4$ to the thickness $d5$. The vertical axis of FIG. 9 indicates the reflected electric field strength normalized by setting the reflected electric field strength in the case where the protective layer **44** is not provided to 1.0.

[0088] The characteristics shown in FIG. 9 were obtained by calculation. Specifically, silver was

used as a constituent material of the metal layer **41** and the metal layer **43** (metal unit **45**), and SiO₂ was used as a constituent material of the dielectric layer **42**. The thickness **d1** was set to 200 nm, the thickness **d2** was set to 40 nm, and the thickness **d3** was set to 40 nm. The width **W1** was set to 120 nm and the width **W2** was set to 300 nm. The length of the metal unit **45** in the X-axis direction was set to 2250 nm. Then, gold, ruthenium, and iridium were used as the constituent materials of the protective layer **44**, the thickness of the protective layer **44** was changed for each constituent material, and the reflected electric field strength at each thickness was calculated.

[0089] As shown in FIG. **9**, it can be seen that the reflected electric field strength decreases as the proportion of the protective layer **44** increases for any of the constituent materials. When the proportion of the protective layer **44** is 20% or less, the reflected electric field strength hardly decreases from the reflected electric field strength in the case where the protective layer **44** is not provided, and the reflected electric field strength is maintained at 90% or more. On the other hand, when the proportion of the protective layer **44** exceeds 20%, the reflected electric field strength sharply decreases, and when the proportion of the protective layer **44** is about 30% to 40% or more, the reflected electric field strength becomes almost zero. From the above, it can be said that as long as the proportion of the protective layer **44** is 20% or less, the intensity of the reflected light **Lr** can be maintained.

[0090] Next, an example of a method for manufacturing the near-eye wearable device **1** will be described with reference to FIGS. **10** to **16**. FIG. **10** is a flowchart showing an example of a method for manufacturing a metasurface reflector. FIG. **11** is a view for explaining the resist film forming step shown in FIG. **10**. FIG. **12** is a view for explaining the pattern forming step shown in FIG. **10**. FIG. **13** is a view for explaining the first protective film forming step shown in FIG. **10**. FIG. **14** is a view for explaining the resist film removal step shown in FIG. **10**. FIG. **15** is a view for explaining the metal film forming step shown in FIG. **10**. FIG. **16** is a view for explaining the metal film removal step shown in FIG. **10**.

[0091] A method **M1** shown in FIG. **10** is a method for manufacturing the metasurface reflector **30**. The method **M1** includes a preparation step **S11**, a resist film forming step **S12**, a pattern forming step **S13**, a first protective film forming step **S14**, and a second protective film forming step **S15**.

<Preparation Step **S11**>

[0092] The preparation step **S11** is a step of preparing a laminate **40**. The laminate **40** is formed by laminating the base material **50**, the metal layer **41**, the dielectric layer **42**, the metal layer **143** (second metal layer) which is a base of the metal layer **43**, and the metal layer **144** (third metal layer) which is a base of the first portion **44a** of the protective layer **44** in sequence in the Z-axis direction. Here, an example in which the metasurface reflector **30** is formed not on the lens **3** but on the base material **50** will be described.

[0093] In the preparation step **S11**, the base material **50** is first prepared, and the base material **50** is set in a vacuum film deposition device. As the base material **50**, a sapphire substrate is exemplified. A flexible sheet may be used as the base material **50**. Then, the metal layer **41** is formed on the surface **50a** of the base material **50**. Specifically, the metal layer **41** is formed by vacuum film deposition using a technique such as a direct current (DC) sputtering. For forming the metal layer **41**, a metal material composed of any metal selected from a group consisting of gold (Au), copper (Cu), silver (Ag), and aluminum (Al) or a metal alloy containing at least one element selected from the above-described group is used. Here, silver is exemplified as a constituent material of the metal layer **41**. The thickness **d1** of the metal layer **41** is, for example, 200 nm.

[0094] In order to enhance the adhesion between the surface **50a** of the base material **50** and the metal layer **41**, the adhesion layer **46** may be formed on the surface **50a**, and the metal layer **41** may be formed on the adhesion layer **46**. The adhesion layer **46** is formed, for example, by sputtering or vapor deposition. For forming the adhesion layer **46**, chromium (Cr) is used, for example. The length (film thickness) of the adhesion layer **46** in the Z-axis direction is, for example, 3 nm.

[0095] Subsequently, the dielectric layer **42** is formed on the metal layer **41**. Specifically, the dielectric layer **42** is formed by vacuum film deposition using a technique such as a radio frequency (RF) sputtering. For forming the dielectric layer **42**, a dielectric material such as silicon dioxide (SiO₂), titanium oxide (TiO₂), magnesium oxide (MgO) or aluminum oxide (Al₂O₃) which can be formed by a semiconductor process is used. Here, silicon dioxide is exemplified as a constituent material of the dielectric layer **42**. The thickness d_2 of the dielectric layer **42** is, for example, 40 nm.

[0096] In order to enhance the adhesion between the metal layer **41** and the dielectric layer **42**, the adhesion layer **47** may be formed on the metal layer **41**, and the dielectric layer **42** may be formed on the adhesion layer **47**. Since the method for forming the adhesion layer **47** is the same as the method for forming the adhesion layer **46**, a detailed description thereof will be omitted. For forming the adhesion layer **47**, chromium (Cr) is used, for example. The length (film thickness) of the adhesion layer **47** in the Z-axis direction is, for example, 3 nm.

[0097] Subsequently, a metal layer **143** is formed on the dielectric layer **42**. Since the method for forming the metal layer **143** is the same as the method for forming the metal layer **41**, a detailed description thereof will be omitted. For forming the metal layer **143**, a metal material made of any metal selected from the group consisting of copper (Cu), silver (Ag), and aluminum (Al) or a metal alloy containing at least one element selected from the above-described group is used. Here, silver is exemplified as a constituent material of the metal layer **143**. The length (thickness d_3) of the metal layer **143** in the Z-axis direction is, for example, 40 nm.

[0098] In order to enhance the adhesion between the dielectric layer **42** and the metal layer **143**, the adhesion layer **148** may be formed on the dielectric layer **42**, and the metal layer **143** may be formed on the adhesion layer **148**. Since the method for forming the adhesion layer **148** is the same as the method for forming the adhesion layer **46**, a detailed description thereof will be omitted. For forming the adhesion layer **148**, chromium (Cr) is used, for example.

[0099] Subsequently, a metal layer **144** is formed on the metal layer **143**. Since the method for forming the metal layer **144** is the same as the method for forming the metal layer **41**, a detailed description thereof will be omitted. For forming the metal layer **144**, a metal material made of any metal selected from the group consisting of gold (Au), ruthenium (Ru), and iridium (Ir), or a metal alloy containing at least one element selected from the above-described group is used. Here, gold is exemplified as a constituent material of the metal layer **144**. The length of the metal layer **144** in the Z-axis direction is, for example, 10 nm. As described above, the laminate **40** is obtained.

<Resist Film Forming Step S12>

[0100] Following the preparation step S11, the resist film forming step S12 is performed. The resist film forming step S12 is a step of forming a resist film **51** (photoresist) on the metal layer **144**. As shown in FIG. 11, in the resist film forming step S12, a liquid resist is applied onto the metal layer **144** using a spin coater or the like, and the applied liquid resist is dried to form the resist film **51**. The thickness of the resist film **51** is, for example, 230 nm.

<Pattern Forming Step S13>

[0101] Following the resist film forming step S12, the pattern forming step S13 is performed. The pattern forming step S13 is a step of forming a pattern **51p** for forming the metal unit **45** constituting the metasurface reflector **30** in the resist film **51**. As shown in FIG. 12, in the pattern forming step S13, the pattern **51p** corresponding to the metal units **45** is transferred onto the resist film **51** using an exposure device such as a KrF exposure device and an electron beam lithography device. Then, the pattern **51p** transferred to the resist film **51** is developed using a developing machine.

<First Protective Film Forming Step S14>

[0102] Following the pattern forming step S13, the first protective film forming step S14 is performed. The first protective film forming step S14 is a step of forming the metal unit **45** in the metal layer **143** and the first portion **44a** (first protective film) covering the top surface **45a** of the

metal unit **45** in the metal layer **144** by milling the metal layer **143** and the metal layer **144** using the pattern **51p**. As shown in FIG. **13**, in the first protective film forming step **S14**, the ion beam is irradiated by the ion milling method until the dielectric layer **42** is exposed. As a result, portions of the metal layer **144**, the metal layer **143**, and the adhesion layer **148** that are not covered with the pattern **51p** are removed, and the first portion **44a**, the metal unit **45**, and the adhesion layer **48** are formed.

<Second Protective Film Forming Step S15>

[0103] Following the first protective film forming step **S14**, the second protective film forming step **S15** is performed. The second protective film forming step **S15** is a step of forming the second portion **44b** (second protective film) on the side surface **45c** of the metal unit **45**. The second protective film forming step **S15** includes a resist film removal step **S16**, a metal film forming step **S17**, and a metal film removal step **S18**.

<Resist Film Removal Step S16>

[0104] The resist film removal step **S16** is a step of removing the resist film **51**. As shown in FIG. **14**, the resist film **51** is removed with an organic solvent (NMP). As a result, a structure **130** is obtained.

<Metal Film Forming Step S17>

[0105] Following the resist film removal step **S16**, the metal film forming step **S17** is performed. The metal film forming step **S17** is a step of forming a metal film **52** on the surface **130a** of the structure **130**. The surface **130a** includes a top surface of the first portion **44a**, a side surface of the first portion **44a**, the side surface **45c** of the metal unit **45**, and a portion of the main surface **42a** of the dielectric layer **42** which is not covered with the metal unit **45**.

[0106] In the metal film forming step **S17**, the metal film **52** is formed by vacuum film deposition using glancing angle sputtering. As shown in FIG. **15**, the rotation axis **AX** of the structure **130** is set in the normal direction of the main surface **42a** of the dielectric layer **42**. The target **T** is arranged such that the metal particles emitted from the target **T** are incident at an incident angle α . In other words, the angle formed by the normal direction of the main surface **Ta** of the target **T** and the normal direction (rotation axis **AX**) of the main surface **42a** of the dielectric layer **42** is set to the incident angle α . The main surface **Ta** is a surface from which metal particles are emitted. The incident angle α is, for example, 67° . Sputtering is performed while the structure **130** is rotated about the rotation axis **AX**. This allows the metal particles to enter between the two metal units **45** adjacent to each other, and the metal film **52** is formed so as to cover the entire surface **130a**.

[0107] For forming the metal film **52**, a metal material made of any metal selected from the group consisting of gold (Au), ruthenium (Ru), and iridium (Ir), or a metal alloy containing at least one element selected from the above-described group is used. Here, gold is exemplified as a constituent material of the metal film **52**. The thickness of the metal film **52** is, for example, 10 nm.

<Metal Film Removal Step S18>

[0108] Following the metal film forming step **S17**, the metal film removal step **S18** is performed. The metal film removal step **S18** is a step of removing the metal film **52** covering the main surface **42a** between the two metal units **45** adjacent to each other. As shown in FIG. **16**, etching (etch-back) is performed in the normal direction of the main surface **42a** by anisotropic etching. The entire surface of the metal film **52** is etched until the main surface **42a** between the two metal units **45** adjacent to each other is exposed. As a result, the metal film **52** covering the top surface of the first portion **44a** and the metal film **52** covering the main surface **42a** between the two metal units **45** adjacent to each other are removed, and a part of the metal film **52** covering the side surface of the first portion **44a** and the side surface **45c** of the metal unit **45** remains, thereby forming the second portion **44b**. Through the above-described etch-back, the second portion **44b** has a shape in which the thickness of the second portion **44b** decreases as the distance from the dielectric layer **42** in the Z-axis direction increases.

[0109] As a result, the metasurface reflector **30** is formed on the surface **50a** of the base material

50.

[0110] Next, another example of the method for manufacturing the near-eye wearable device **1** will be described with reference to FIGS. **17** to **23**. FIG. **17** is a flowchart showing another example of a method for manufacturing a metasurface reflector. FIG. **18** is a view for explaining the resist film forming step shown in FIG. **17**. FIG. **19** is a view for explaining the pattern forming step shown in FIG. **17**. FIG. **20** is a view for explaining the first protective film forming step shown in FIG. **17**. FIG. **21** is a view for explaining the metal film forming step shown in FIG. **17**. FIG. **22** is a view for explaining the metal film removal step shown in FIG. **17**. FIG. **23** is a view for explaining the resist film removal step shown in FIG. **17**.

[0111] A method M2 shown in FIG. **17** is a method for manufacturing the metasurface reflector **30**. The method M2 includes a preparation step S21, a resist film forming step S22, a pattern forming step S23, a first protective film forming step S24, and a second protective film forming step S25.

<Preparation Step S21>

[0112] The preparation step S21 is a step of preparing the laminate **40**. Since the preparation step S21 is the same as the preparation step S11, a detailed description thereof will be omitted.

<Resist Film Forming Step S22>

[0113] Following the preparation step S21, the resist film forming step S22 is performed. The resist film forming step S22 is a step of forming a multilayer resist film **55** including a lower resist film **53** and an upper resist film **54** on the metal layer **144**. As shown in FIG. **18**, in the resist film forming step S22, a liquid resist is applied onto the metal layer **144** using a spin coater or the like, and the applied liquid resist is dried to form the lower resist film **53**. The thickness of the lower resist film **53** is, for example, 38 nm. Further, a liquid resist is applied onto the lower resist film **53** using a spin coater or the like, and the applied liquid resist is dried to form the upper resist film **54**. The thickness of the upper resist film **54** is, for example, 250 nm.

<Pattern Forming Step S23>

[0114] Following the resist film forming step S22, the pattern forming step S23 is performed. The pattern formation step S23 is a step of forming a pattern **55p** for forming the metal unit **45** constituting the metasurface reflector **30** in the multilayer resist film **55**. As shown in FIG. **19**, in the pattern forming step S23, the pattern **55p** corresponding to the metal units **45** is transferred onto the upper resist film **54** using an exposure device such as a KrF exposure device and an electron beam lithography device. Then, the pattern **55p** transferred to the upper resist film **54** is developed using a developing machine. At this time, the lower resist film **53** is isotropically dissolved without causing a photoreaction, and an overhang is formed in which the lower resist film **53** becomes one size smaller than the upper resist film **54** when viewed from the laminating direction. In other words, the edge of the lower resist film **53** is located inside the side surface of the upper resist film **54**. The position of the edge of the lower resist film **53** is adjusted by the time of the development process.

<First Protective Film Forming Step S24>

[0115] Following the pattern forming step S23, the first protective film forming step S24 is performed. The first protective film forming step S24 is a step of forming the metal unit **45** in the metal layer **143** and the first portion **44a** (first protective film) covering the top surface **45a** of the metal unit **45** in the metal layer **144** by milling the metal layer **143** and the metal layer **144** using the pattern **55p**. As shown in FIG. **20**, in the first protective film forming step S24, the ion beam is irradiated by the ion milling method until the dielectric layer **42** is exposed. As a result, portions of the metal layer **144**, the metal layer **143**, and the adhesion layer **148** that are not covered with the pattern **55p** are removed, and the first portion **44a**, the metal unit **45**, and the adhesion layer **48** are formed. As a result, a structure **230** is obtained.

<Second Protective Film Forming Step S25>

[0116] Following the first protective film forming step S24, the second protective film forming step S25 is performed. The second protective film forming step S25 is a step of forming the second

portion **44b** (second protective film) on the side surface **45c** of the metal unit **45**. The second protective film forming step **S25** includes a metal film forming step **S26**, a metal film removal step **S27**, and a resist film removal step **S28**.

<Metal Film Forming Step S26>

[0117] The metal film forming step **S26** is a step of forming a metal film **56** on the surface **230a** of the structure **230**. The surface **230a** includes a top surface of the upper resist film **54**, a side surface of the upper resist film **54**, a side surface of the first portion **44a**, the side surface **45c** of the metal unit **45**, and a portion of the main surface **42a** of the dielectric layer **42** which is not covered with the metal unit **45**.

[0118] In the metal film forming step **S26**, the metal film **56** is formed by vacuum film deposition using glancing angle sputtering. As shown in FIG. **21**, the rotation axis **AX** of the structure **230** is set in the normal direction of the main surface **42a** of the dielectric layer **42**. The target **T** is arranged such that the metal particles emitted from the target **T** are incident at an incident angle β . That is, the angle formed by the normal direction of the main surface **Ta** of the target **T** and the normal direction (rotation axis **AX**) of the main surface **42a** of the dielectric layer **42** is set to the incident angle β . The incident angle β is, for example, 0° to 28° . Sputtering is performed while the structure **230** is rotated about the rotation axis **AX**. This allows the metal particles to enter between the two metal units **45** adjacent to each other.

[0119] Since the edge of the lower resist film **53** is located inside the side surface of the upper resist film **54**, the metal film **56** is not formed on the edge of the lower resist film **53**. Accordingly, the metal film **56** is formed on the top surface and the side surface of the upper resist film **54**, the side surface of the first portion **44a**, the side surface **45c** of the metal unit **45**, and the portion of the main surface **42a** of the dielectric layer **42** which is not covered with the metal unit **45**. A gap **56a** is formed in the metal film **56** along the edge of the lower resist film **53**.

[0120] For forming the metal film **56**, a metal material made of any metal selected from the group consisting of gold (Au), ruthenium (Ru), and iridium (Ir), or a metal alloy containing at least one element selected from the above-described group is used. Here, gold is exemplified as a constituent material of the metal film **56**. The thickness of the metal film **56** is, for example, 10 nm.

<Metal Film Removal Step S27>

[0121] Following the metal film forming step **S26**, the metal film removal step **S27** is performed. The metal film removal step **S27** is a step of removing the metal film **56** covering the main surface **42a** between the two metal units **45** adjacent to each other. As shown in FIG. **22**, in the metal film removal step **S27**, the entire surface of the metal film **56** is irradiated with the ion beam by the ion milling method until the main surface **42a** between the two metal units **45** adjacent to each other is exposed. As a result, the metal film **56** covering the top surface of the upper resist film **54** and the metal film **56** covering the main surface **42a** between the two metal units **45** adjacent to each other are removed, and the metal film **56** covering the side surface of the first portion **44a** and the side surface **45c** of the metal unit **45** remains, thereby forming the second portion **44b**. The metal film **56** covering the side surface of the upper resist film **54** also remains.

<Resist Film Removal Step S28>

[0122] Following the metal film removal step **S27**, the resist film removal step **S28** is performed. The resist film removal step **S28** is a step of removing the multilayer resist film **55**. As shown in FIG. **23**, the organic solvent (NVP) enters the gap **56a** and reaches the edge of the lower resist film **53**, thereby dissolving the lower resist film **53**. As a result, the metal film **56** covering the side surface of the upper resist film **54** and the multilayer resist film **55** are removed.

[0123] As a result, the metasurface reflector **30** is formed on the surface **50a** of the base material **50**.

[0124] According to the method **M1** or the method **M2** described above, a plurality of metasurface reflectors **30** are formed on one base material **50**. Thus, by cutting the base material **50**, a portion including one metasurface reflector **30** is obtained. The metasurface reflector **30** is formed on the

inner surface **3a** of the lens **3** by attaching the portion of the base material **50** to a predetermined area of the inner surface **3a** of the lens **3**.

[0125] Subsequently, the frame **2** on which the optical engine **20** is mounted is prepared, and the lens **3** on which the metasurface reflector **30** is formed is mounted on the rim **2a** of the frame **2**. As described above, the near-eye wearable device **1** is manufactured.

[0126] The metasurface reflector **30** may be formed directly on the inner surface **3a** of the lens **3**. The method for forming the metasurface reflector **30** on the inner surface **3a** of the lens **3** is the same as the method for forming the metasurface reflector **30** on the surface **50a** of the base material **50**. In this case, the metasurface reflector **30** is formed in a desired region on the inner surface **3a**.

[0127] In the near-eye wearable device **1**, the projection device **10**, and the metasurface reflector **30** described above, the metasurface reflector **30** is divided into the plurality of unit regions **31** arranged in the X-axis direction along the main surface **42a** and in the Y-axis direction along the main surface **42a** and intersecting (orthogonal to) the X-axis direction, the metal unit **45** is provided in each unit region **31**, and the protective layer **44** covers the top surface **45a** and the side surface **45c** of the metal unit **45**. Accordingly, since the top surface **45a** and the side surface **45c** of the metal unit **45** are not exposed to the air when the near-eye wearable device **1**, the projection device **10**, and the metasurface reflector **30** are used, the possibility that the metal unit **45** is oxidized and sulfurized is reduced. Further, since the protective layer **44** is made of a metal having a standard electrode potential higher than that of the metal constituting the metal layer **43**, the protective layer **44** is less likely to be oxidized and sulfurized than the metal layer **43**. Accordingly, in the near-eye wearable device **1**, the projection device **10**, and the metasurface reflector **30**, the optical characteristics hardly change. As a result, deterioration of the reflection characteristics can be suppressed.

[0128] As the ratio of the thickness **d4** to the thickness **d5** increases, the influence of the protective layer **44** on the reflection characteristics of the metasurface reflector **30** increases. For example, gold (Au) reflects red light but easily absorbs green light and blue light. Ruthenium and iridium easily absorb light in the visible light region. Accordingly, if the ratio of the thickness **d4** to the thickness **d5** is high, the reflection characteristics of the metasurface reflector **30** may be deteriorated. On the other hand, in the metasurface reflector **30**, the thickness **d4** is 20% or less of the thickness **d5**. Therefore, the protective layer **44** can be provided while suppressing the influence of the protective layer **44** on the reflection characteristics of the metasurface reflector **30**. Accordingly, deterioration of the reflection characteristics can be further suppressed.

[0129] Metals such as molybdenum, titanium, and tungsten also have high resistance to oxidation and sulfidation and exhibit high corrosion resistance, but have large absorption coefficients in the visible light region. Accordingly, if the protective layer **44** is made of any of these metals, the reflection characteristics of the metasurface reflector **30** may be deteriorated. The absorption coefficients of gold, ruthenium, and iridium in the visible light region are smaller than the absorption coefficients of metals such as molybdenum, titanium, and tungsten in the visible light region. Accordingly, when the protective layer **44** is made of a metal containing at least one element selected from the group consisting of gold, ruthenium, and iridium, the deterioration of the reflection characteristics of the metasurface reflector **30** can be further suppressed.

[0130] The metal unit **45** is a metal body having a trapezoidal shape when viewed from the Z-axis direction. Therefore, the structure of the metal unit **45** can be simplified as compared with the case where the metal unit is composed of a plurality of metal bodies. Accordingly, the manufacturing of the metasurface reflector **30** can be facilitated.

[0131] When the length of the metal unit **45** in the X-axis direction is 500 nm or more and 2500 nm or less, the reflected light **Lr** corresponding to a viewing angle of 40° to 60° can be obtained. When the thickness **d3** is 10 nm or more and 100 nm or less, the electromagnetic resonance with the metal layer **41** through the dielectric layer **42** is effectively generated, and the reflected light **Lr** having a strong electric field strength can be obtained. When the width **W1** is 10 nm or more and 200 nm or

less and the width **W2** is 100 nm or more and 500 nm or less, the phase difference between the phase of the reflected light **Lr** at the width **W1** and the phase of the reflected light **Lr** at the width **W2** can be set to 360° (2π radians) for the laser light **Ls** of the visible light. As described above, by setting the respective dimensions of the metal unit **45** within the above-described ranges, the reflection efficiency for visible light can be enhanced.

[0132] The metal layer **43** is made of a metal containing at least one element selected from the group consisting of silver, aluminum, and copper. Therefore, the metal layer **43** having a relatively high reflectance in the visible light region and a relatively high electric conductivity is obtained. Accordingly, the electromagnetic resonance between the metal layer **41** and the metal layer **43** can be strengthened, and the reflection efficiency can be enhanced.

[0133] The dielectric layer **42** is made of a material transparent in the visible light region. Accordingly, the absorption rate of visible light in the dielectric layer **42** is suppressed, so that the reflection efficiency for visible light can be enhanced.

[0134] The dielectric layer **42** is made of, for example, one compound selected from the group consisting of silicon oxide, titanium oxide, magnesium oxide, and aluminum oxide. In this case, the dielectric layer **42** having a dielectric constant that does not interfere with the electromagnetic action is obtained. Accordingly, the electromagnetic resonance between the metal layer **41** and the metal layer **43** can be strengthened, and the reflection efficiency can be enhanced.

[0135] The thickness **d1** is, for example, 50 nm or more and 1000 nm or less. The thickness **d2** is, for example, 10 nm or more and 100 nm or less. In this case, the possibility that the dielectric layer **42** interferes with the electromagnetic action can be reduced, and the possibility that the laser light **Ls** passes through the metal layer **41** can be reduced. Accordingly, the reflection efficiency can be enhanced.

[0136] The thinner the second portion **44b** is, the higher the reflected electric field strength becomes. Therefore, when the protective layer **44** is configured such that the thickness of the second portion **44b** decrease as the distance from the dielectric layer **42** in the Z-axis direction increases, the protective layer **44** can be provided while suppressing the influence on the reflection characteristics of the metasurface reflector **30**. Therefore, the deterioration of the reflection characteristics can be further suppressed.

[0137] In the method **M1** for manufacturing the metasurface reflector **30**, the pattern **51p** is formed in the resist film **51** formed on the metal layer **144** of the laminate **40**, and the metal unit **45** and the first portion **44a** covering the top surface **45a** of the metal unit **45** are formed by milling the metal layer **143** and the metal layer **144** using the pattern **51p**. In the method **M2** for manufacturing the metasurface reflector **30**, the pattern **55p** is formed in the multilayer resist film **55** formed on the metal layer **144** of the laminate **40**, and the metal unit **45** and the first portion **44a** covering the top surface **45a** of the metal unit **45** are formed by milling the metal layer **143** and the metal layer **144** using the pattern **55p**. Further, the second portion **44b** is formed on the side surface **45c** of the metal unit **45**. In the metasurface reflector **30** manufactured in this manner, since the top surface **45a** and the side surface **45c** of the metal unit **45** are not exposed to the air when the metasurface reflector **30** is used, the possibility that the metal unit **45** is oxidized and sulfurized is reduced. Furthermore, since the protective layer **44** is made of a metal having a standard electrode potential higher than that of the metal constituting the metal layer **43**, the protective layer **44** is less likely to be oxidized and sulfurized than the metal layer **43**. Accordingly, in the metasurface reflector **30**, the optical characteristics hardly change. As a result, deterioration of the reflection characteristics can be suppressed.

[0138] As described above, in the manufacturing process of the metasurface reflector **30**, the resist film is removed with an organic solvent (NMP). At this time, if the metal layer **43** (metal unit **45**) is exposed, the metal layer **43** may be corroded by the organic solvent. On the other hand, in the methods **M1** and **M2**, the resist films **51** and **55** are removed while the top surface **45a** of the metal unit **45** is covered with the first portion **44a** of the protective layer **44**. Since the first portion **44a** is

made of a metal having a standard electrode potential higher than that of the metal constituting the metal layer **43**, the first portion **44a** is less susceptible to corrosion than the metal layer **43**. Accordingly, in the manufacturing process of the metasurface reflector **30**, the possibility that the metal layer **43** (metal unit **45**) is corroded can be reduced, and the corrosion resistance can be improved.

[0139] In the method M1, the metal film **52** is formed on the surface **130a** of the structure **130** obtained by removing the resist film **51**, and the second portion **44b** is formed by removing the portion of the metal film **52** formed on the main surface **42a** of the dielectric layer **42**. Thus, the second portion **44b** can be formed on the side surface **45c** of the metal unit **45** by a relatively simple process.

[0140] In the method M2, the multilayer resist film **55** is removed in a state where the top surface **45a** of the metal unit **45** is covered with the first portion **44a** and the side surface **45c** of the metal unit **45** is covered with the second portion **44b**. Therefore, since the top surface **45a** and the side surface **45c** of the metal unit **45** are not exposed to the organic solvent used for removing the multilayer resist film **55**, the possibility that the metal unit **45** is corroded is reduced. As a result, deterioration of the reflection characteristics of the metasurface reflector **30** can be further suppressed.

[0141] In the metal film forming step S17, the metal film **52** is formed using glancing angle sputtering. As a result, the metal particles enter between the two metal units **45** adjacent to each other, and the metal film **52** is easily formed on the side surface **45c** of the metal unit **45**. Similarly, in the metal film forming step S26, the metal film **56** is formed using glancing angle sputtering. As a result, the metal particles enter between the two metal units **45** adjacent to each other, and the metal film **56** is easily formed on the side surface **45c** of the metal unit **45**. Therefore, the time required for manufacturing the metasurface reflector **30** can be reduced.

[0142] The metasurface reflector, the projection device, the near-eye wearable device, and the method for manufacturing the metasurface reflector according to the present disclosure are not limited to the above embodiments.

[0143] The near-eye wearable device **1** may be virtual reality (VR) glasses.

[0144] In the above-described embodiments, the projection device **10** directly projects (draws) an image onto the retina RE of the user of the near-eye wearable device **1**. The projection device **10** may project an image onto the metasurface reflector **30**.

[0145] The metasurface reflector **30** may be applied to devices other than the projection device **10**. For example, the metasurface reflector **30** may be applied to a general image projection surface such as an image screen.

[0146] The plurality of unit regions **31** may include a unit region **31** for a red component, a unit region **31** for a green component, and a unit region **31** for a blue component. In the X-axis direction, the unit region **31** for the red component, the unit region **31** for the green component, and the unit region **31** for the blue component may be repeatedly arranged in that order. In the Y-axis direction as well, the unit region **31** for the red component, the unit region **31** for the green component, and the unit region **31** for the blue component may be repeatedly arranged in that order.

[0147] In the above-described embodiments, the metal unit **45** is provided in each of all the unit regions **31** among the plurality of unit regions **31**, but the metal unit **45** may be provided in each of some of the unit regions **31** among the plurality of unit regions **31**.

[0148] The method for determining the length L_x is not limited to the method described in the above-described embodiments. For example, the lengths L_x of the unit regions **31** located at both ends of the metasurface reflector **30** in the X-axis direction may be determined by the above-described method, and the lengths L_x of the unit regions **31** located therebetween may be determined so as to gradually change from the length L_x of the unit region **31** located at one end of the metasurface reflector **30** in the X-axis direction to the length L_x of the unit region **31** located at the other end thereof.

[0149] The metal unit **45** is not limited to one metal body having a trapezoidal shape, and may be composed of, for example, a plurality of metal bodies arranged in the X-axis direction.

Additional Statements

Clause 1

[0150] A metasurface reflector comprising: [0151] a first metal layer and a second metal layer stacked in a first direction; [0152] a dielectric layer provided between the first metal layer and the second metal layer in the first direction; and [0153] a protective layer covering the second metal layer, [0154] wherein the dielectric layer includes a main surface on which the second metal layer is provided, [0155] wherein the metasurface reflector is divided into a plurality of unit regions arranged in a second direction along the main surface and in a third direction along the main surface and intersecting the second direction, [0156] wherein the second metal layer includes metal units respectively provided in all or some of the plurality of unit regions [0157] wherein the protective layer is made of a metal having a standard electrode potential higher than that of a metal constituting the second metal layer, [0158] wherein each of the metal units includes a bottom surface facing the dielectric layer in the first direction, a top surface provided opposite to the bottom surface in the first direction, and a side surface connecting the top surface and the bottom surface, and [0159] wherein the protective layer includes a first portion covering the top surface and a second portion covering the side surface.

Clause 2

[0160] The metasurface reflector according to clause 1, [0161] wherein a thickness of the protective layer is 20% or less of a sum of a length of the second metal layer in the first direction and the thickness of the protective layer.

Clause 3

[0162] The metasurface reflector according to clause 1 or 2, [0163] wherein each of the metal units is a metal body having a trapezoidal shape when viewed from the first direction.

Clause 4

[0164] The metasurface reflector according to clause 3, [0165] wherein a length of the metal body in the second direction is 500 nm or more and 2500 nm or less, [0166] wherein a length of the metal body in the first direction is 10 nm or more and 100 nm or less, [0167] wherein a length of a short side of the metal body is 10 nm or more and 200 nm or less, and [0168] wherein a length of a long side of the metal body is larger than the length of the short side and is 100 nm or more and 500 nm or less.

Clause 5

[0169] The metasurface reflector according to any one of clauses 1 to 4, [0170] wherein the protective layer is made of a metal containing at least one element selected from a group consisting of gold, ruthenium, and iridium.

Clause 6

[0171] The metasurface reflector according to any one of clauses 1 to 5, [0172] wherein the second metal layer is made of a metal containing at least one element selected from a group consisting of silver, aluminum, and copper.

Clause 7

[0173] The metasurface reflector according to any one of clauses 1 to 6, [0174] wherein the dielectric layer is made of a material transparent in a visible light region.

Clause 8

[0175] The metasurface reflector according to clause 7, [0176] wherein the dielectric layer is made of a compound selected from a group consisting of silicon oxide, titanium oxide, magnesium oxide, and aluminum oxide.

Clause 9

[0177] The metasurface reflector according to any one of clauses 1 to 8, [0178] wherein a length of the dielectric layer in the first direction is 10 nm or more and 100 nm or less, and [0179] wherein a

length of the first metal layer in the first direction is 50 nm or more and 1000 nm or less.

Clause 10

[0180] The metasurface reflector according to any one of clauses 1 to 9, [0181] wherein a thickness of the second portion decreases as a distance from the dielectric layer increases in the first direction.

Clause 11

[0182] A projection device mounted on a near-eye wearable device, the projection device comprising: [0183] a light source configured to emit laser light; [0184] a movable mirror configured to perform scanning with the laser light; and [0185] the metasurface reflector according to any one of clauses 1 to 10, the metasurface reflector configured to reflect the laser light that has passed through the movable mirror to cause a user wearing the near-eye wearable device to visually recognize an image.

Clause 12

[0186] A near-eye wearable device comprising: [0187] the projection device according to clause 11; and [0188] a lens provided with the metasurface reflector.

Clause 13

[0189] A method for manufacturing a metasurface reflector, the method comprising: [0190] a step of preparing a laminate in which a first metal layer, a dielectric layer, a second metal layer, and a third metal layer are sequentially laminated in a first direction; [0191] a step of forming a resist film on the third metal layer; [0192] a step of forming, on the resist film, a pattern for forming a metal unit constituting a metasurface reflector, the metal unit including a bottom surface facing the dielectric layer in the first direction, a top surface provided opposite to the bottom surface in the first direction, and a side surface connecting the top surface and the bottom surface; [0193] a step of forming the metal unit and a first protective film covering the top surface by milling the second metal layer and the third metal layer using the pattern; and [0194] a step of forming a second protective film covering the side surface of the metal unit, [0195] wherein the first protective film and the second protective film are made of a metal having a standard electrode potential higher than that of a metal constituting the second metal layer.

Clause 14

[0196] The method for manufacturing a metasurface reflector according to clause 13, [0197] wherein the step of forming the second protective film includes: [0198] a step of removing the resist film; [0199] a step of forming a metal film on a surface of a structure obtained by removing the resist film; and [0200] a step of forming the second protective film by removing a portion of the metal film formed on a main surface of the dielectric layer.

Clause 15

[0201] The method for manufacturing a metasurface reflector according to clause 13, [0202] wherein in the step of forming the resist film, a multilayer resist film including a lower resist film and an upper resist film is formed on the third metal layer, and [0203] the step of forming the second protective film includes: [0204] a step of forming a metal film on a surface of a structure obtained by the step of forming the first protective film; [0205] a step of forming the second protective film by removing a portion of the metal film formed on a main surface of the dielectric layer; and [0206] a step of removing the multilayer resist film after the second protective film is formed.

Clause 16

[0207] The method for manufacturing a metasurface reflector according to clause 14 or 15, [0208] wherein in the step of forming the metal film, the metal film is formed using glancing angle sputtering.

Claims

- 1.** A metasurface reflector comprising: a first metal layer and a second metal layer stacked in a first direction; a dielectric layer provided between the first metal layer and the second metal layer in the first direction; and a protective layer covering the second metal layer, wherein the dielectric layer includes a main surface on which the second metal layer is provided, wherein the metasurface reflector is divided into a plurality of unit regions arranged in a second direction along the main surface and in a third direction along the main surface and intersecting the second direction, wherein the second metal layer includes metal units respectively provided in all or some of the plurality of unit regions wherein the protective layer is made of a metal having a standard electrode potential higher than that of a metal constituting the second metal layer, wherein each of the metal units includes a bottom surface facing the dielectric layer in the first direction, a top surface provided opposite to the bottom surface in the first direction, and a side surface connecting the top surface and the bottom surface, and wherein the protective layer includes a first portion covering the top surface and a second portion covering the side surface.
- 2.** The metasurface reflector according to claim 1, wherein a thickness of the protective layer is 20% or less of a sum of a length of the second metal layer in the first direction and the thickness of the protective layer.
- 3.** The metasurface reflector according to claim 1, wherein each of the metal units is a metal body having a trapezoidal shape when viewed from the first direction.
- 4.** The metasurface reflector according to claim 3, wherein a length of the metal body in the second direction is 500 nm or more and 2500 nm or less, wherein a length of the metal body in the first direction is 10 nm or more and 100 nm or less, wherein a length of a short side of the metal body is 10 nm or more and 200 nm or less, and wherein a length of a long side of the metal body is larger than the length of the short side and is 100 nm or more and 500 nm or less.
- 5.** The metasurface reflector according to claim 1, wherein the protective layer is made of a metal containing at least one element selected from a group consisting of gold, ruthenium, and iridium.
- 6.** The metasurface reflector according to claim 1, wherein the second metal layer is made of a metal containing at least one element selected from a group consisting of silver, aluminum, and copper.
- 7.** The metasurface reflector according to claim 1, wherein the dielectric layer is made of a material transparent in a visible light region.
- 8.** The metasurface reflector according to claim 7, wherein the dielectric layer is made of a compound selected from a group consisting of silicon oxide, titanium oxide, magnesium oxide, and aluminum oxide.
- 9.** The metasurface reflector according to claim 1, wherein a length of the dielectric layer in the first direction is 10 nm or more and 100 nm or less, and wherein a length of the first metal layer in the first direction is 50 nm or more and 1000 nm or less.
- 10.** The metasurface reflector according to claim 1, wherein a thickness of the second portion decreases as a distance from the dielectric layer increases in the first direction.
- 11.** A projection device mounted on a near-eye wearable device, the projection device comprising: a light source configured to emit laser light; a movable mirror configured to perform scanning with the laser light; and the metasurface reflector according to claim 1, the metasurface reflector configured to reflect the laser light that has passed through the movable mirror to cause a user wearing the near-eye wearable device to visually recognize an image.
- 12.** A near-eye wearable device comprising: the projection device according to claim 11; and a lens provided with the metasurface reflector.
- 13.** A method for manufacturing a metasurface reflector, the method comprising: a step of preparing a laminate in which a first metal layer, a dielectric layer, a second metal layer, and a third metal layer are sequentially laminated in a first direction; a step of forming a resist film on the third metal layer; a step of forming, on the resist film, a pattern for forming a metal unit constituting a

metasurface reflector, the metal unit including a bottom surface facing the dielectric layer in the first direction, a top surface provided opposite to the bottom surface in the first direction, and a side surface connecting the top surface and the bottom surface; a step of forming the metal unit and a first protective film covering the top surface by milling the second metal layer and the third metal layer using the pattern; and a step of forming a second protective film covering the side surface of the metal unit, wherein the first protective film and the second protective film are made of a metal having a standard electrode potential higher than that of a metal constituting the second metal layer.

14. The method for manufacturing a metasurface reflector according to claim 13, wherein the step of forming the second protective film includes: a step of removing the resist film; a step of forming a metal film on a surface of a structure obtained by removing the resist film; and a step of forming the second protective film by removing a portion of the metal film formed on a main surface of the dielectric layer.

15. The method for manufacturing a metasurface reflector according to claim 13, wherein in the step of forming the resist film, a multilayer resist film including a lower resist film and an upper resist film is formed on the third metal layer, and the step of forming the second protective film includes: a step of forming a metal film on a surface of a structure obtained by the step of forming the first protective film; a step of forming the second protective film by removing a portion of the metal film formed on a main surface of the dielectric layer; and a step of removing the multilayer resist film after the second protective film is formed.

16. The method for manufacturing a metasurface reflector according to claim 14, wherein in the step of forming the metal film, the metal film is formed using glancing angle sputtering.
